

William Dong

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EDUCATION

Carnegie Mellon University B.S. Mechanical Engineering, Incoming Freshman	Expected May 2030
Great Valley High School Senior · Malvern, PA · GPA 4.0 / 4.0 Relevant coursework: AP Physics C (Mechanics, E&M), AP Calculus BC, AP CSA	2022 – 2026

EXPERIENCE

MechE Intern, UGV Defense Company <i>Raptor Defense Company</i> - Led SCORPION, an expendable payload vehicle, as lead and sole mechanical engineer, owning the full mechanical design from concept to working prototype. - Designed the complete assembly in Onshape in 6 days, then manufactured and assembled it in 3. - Delivered on spec: exceeded 50 mph (conservative estimate), used under 50% of the R&D budget, and passed the preliminary test matrix. - Built a Portable Control Unit for the electronics team in SolidWorks.	Summer 2026
Founder & CEO <i>Formula Theory & Automotive Design (FTAD) · 501(c)(3) Nonprofit</i> - Founded an F1-inspired robotics competition nonprofit, securing \$65,000+ in funding through targeted sponsorships and grant acquisition. - Conducted extensive market research and obtained international product validation, establishing FTAD's competition framework across multiple countries. - Engineered competition vehicle designs in Onshape and Fusion 360, applying CFD simulation and topology optimization to maximize aerodynamic and structural performance.	2024 – Present
Independent Engineering Research <i>Carnegie Mellon University</i> - Designing and prototyping a custom cycloidal drive, studying reduction mechanisms, load paths, backlash, and precision manufacturing. - Pending formal-mentorship requests with several Robotics Institute (RI) faculty.	2026 – Present
Design Lead & CAD Specialist <i>VEX Robotics · World-Qualified Competitive Team</i> Led all mechanical design and CAD work on a team that qualified for the VEX World Championship in the 2024 season.	2024 – 2026

LEADERSHIP & ACTIVITIES

Co-Founder <i>NextGen Business & Law Forum · 501(c)(3) Nonprofit</i> - Organized entrepreneurship competitions for 500+ students across the United States. - Oversaw event logistics, curriculum development, and partner outreach.	2022 – 2024
Future Business Leaders of America (FBLA) <i>Web Development · National Qualifier · 4th in Pennsylvania</i> Learned web-dev logic in 4 months and built the competition project in a team of 3.	2025
Competitive Boxer <i>USA Boxing</i> - Two-time PA State Champion (2024 & 2025). - Volunteer boxing coach, mentoring athletes ages 12–34 in technique, discipline, and sportsmanship.	2023 – Present

SKILLS

CAD & Design: Onshape, Fusion 360, SolidWorks, CFD Simulation, Topology Optimization
Programming: Python, Java, C++
Engineering: Kinematic Systems, Structural Analysis, Aerodynamic Simulation, Rapid Prototyping, UGV Systems

INTERESTS

Kinematics engineering, automotive engineering & racing technology, unmanned systems, defense technology, entrepreneurship, and combat sports.